

PRIVATE PLACEMENT MEMORANDUM / KEY INVESTOR INFORMATION

AI-Sangmoo Dynamic Alpha Hedge Strategy

(C1-M2 Quantitative Engine)

Field	Specification
Document Type	Prospectus / PPM (Private Placement Memorandum) & KIID-style Key Facts
Strategy Code	C1-M2
Classification	Quantitative Systematic — Trend-Following & Dynamic Regime Overlay
Base Currency	United States Dollar (USD)
Valuation Frequency	Daily (mark-to-market, US equity session close)
Document Date	8 September 2026
Backtest Audit Cut-off	28 August 2026
Status	Research / Simulated Track Record — Not an offer to the public

IMPORTANT: This document describes a *systematic trading strategy* and its audited historical simulation. It is **not** a registered collective investment scheme prospectus under the U.S. Securities Act, the EU UCITS/AIFMD frameworks, or the Korean FSC public-offering rules, unless and until separately registered. Distribution is limited to **qualified / accredited / professional investors** where permitted by local law.

I. Executive Summary

1.1 Fund / Strategy Identity

Item	Detail
Legal / Marketing Name	AI-Sangmoo Dynamic Alpha Hedge Strategy (C1-M2 Engine)
Strategy Type	Quantitative Systematic Trend-Following & Dynamic Regime Overlay
Reference Benchmarks	NASDAQ-100 (QQQ); S&P 500 (SPY); secondary ceiling check vs ProShares Ultra QQQ (QLD , 2×)
Target Investor	Long-horizon capital appreciators seeking asymmetric payoff and measurable excess return (<i>alpha</i>) versus NASDAQ-100, comfortable with equity-like drawdowns
Recommended Capital Plan	Seed USD 8,000 + USD 400 on each month's first U.S. trading session (<i>DCA / MWRR</i>)
Minimum Notionals (operational)	Live KIS brokerage account with sufficient buying power for 1.5× Nasdaq exposure in low-vol bull regimes

1.2 One-Sentence Thesis

Convert AI-Sangmoo's 17-year proprietary swing constitution (Ichimoku *Kijun* life-line, volume-dry-up, dual-momentum leadership) into a **fully mechanical**, point-in-time, friction-aware book that **never parks idle cash**, scales conviction 50/30/20 into Composite-RS leaders, and applies a **−5.0% hard stop** with uncapped trailing winners — the C1-M2 engine.

1.3 Headline Simulated Results (Audited Friction)

Friction stack applied uniformly: **next-session open fills, 10 bps slippage, 8 bps commission, Point-in-Time (PIT) dynamic 60-name universe.**

Mandate	Window	Net ROI / CAGR	Max Drawdown	Notes
M2 Lump-sum	2018-08-31 → 2026-08-28	CAGR 22.00%	-51.13%	Beats QQQ B&H CAGR 18.39%
C1 Lump-sum	same	CAGR 18.47%	-41.71%	Baseline champion pre-stop widen
M2 DCA (\$8k + \$400/mo)	same	ROI +276.5% / XIRR 27.48%	-48.15%	Final \$174,683 on \$46,400 invested
QQQ DCA (identical flows)	same	ROI +155.8% / XIRR 19.49%	-28.90%	Final \$118,705 — M2 excess ≈ +\$55,979

II. Investment Philosophy & Strategy

2.1 Core Philosophy

- Codified Discretion → Systematic Rules**
The discretionary framework of Al-Sangmoo (Oh Dong-seok): 17 years of prop/hedge-fund practice distilled into a **3-Gate** entry model — Engine A (20-day breakout above cloud), Engine B (*Kijun* pullback reclaim), Gate VDU (volume dry-up compression).
- Asymmetric Payoff (*Let Winners Run*)**
Accept a high rate of small, mechanical losses (hard stop) in exchange for **uncapped right-tail** capture via Ichimoku/ATR trailing after a +15% peak trigger. Win rate is intentionally modest; **Profit Factor** and **Calmar** are the primary quality metrics.
- No Cash Drag**
Empty conviction slots do **not** accrue idle cash. Residual capital is continuously allocated to a **QQQ cash-proxy overlay**, with conditional QLD mix when gross leverage targets 1.5×.

2.2 Dual-Alpha Architecture

C1-M2 PORTFOLIO NAV

SATELLITE ALPHA	BASE BETA (Cash Proxy)
1-3 leadership slots	Idle residual → QQQ (+ QLD)
Composite RS ranked	Regime-adaptive gross exposure
PIT Dynamic-60 pool	Never 50% cash under SMA200*

* Prior experiments showed SMA200 cash sleeves underperform QQQ B&H.

Sleeve	Role	Selection
Base Beta	Structural Nasdaq exposure; eliminates cash drag	Always-on QQQ; QLD overlay only when leverage target > 1.0×
Satellite Alpha	Idiosyncratic leadership premium	Top Composite RS names that beat QQQ's own Composite RS (dual-momentum gate) inside the PIT 11-sector Dynamic-60

Composite Relative Strength (ranking score):

[

$$\text{CompositeRS}_t = 0.40 \cdot RS_{\{21\}} + 0.35 \cdot RS_{\{63\}} + 0.25 \cdot RS_{\{126\}}$$

]

2.3 Dynamic Leverage Module

Regime Condition	Gross Nasdaq Target	Implementation
SPY ≥ SMA(200) and VIX < 20	1.5×	Mix residual cash into QQQ (1×) + QLD (2×) without exceeding spendable idle capital
Otherwise (incl. SPY < SMA(200) or VIX ≥ 20)	1.0×	QLD off; QQQ core retained

Leverage is **not** synthetic margin beyond cash; when fully invested in equities, gross exposure cannot rise further.

III. Portfolio Construction & Allocation

3.1 Three-Slot Conviction Matrix (Bull Regime)

Slot	Rank by Composite RS	Target Weight
Slot 1	#1 leadership name	50%
Slot 2	#2 leadership name	30%
Slot 3	#3 leadership name	20%

Bear regime (SPY < SMA200): new entries capped at **2 slots × 25%** (Baseline C1). Residual weight remains in the QQQ proxy. *(Tuning variant M2 historically also tested a soft-bear 2×40% sleeve; production PPM risk section uses the −5% stop as the defining M2 control.)*

3.2 Rebalancing & Execution Protocol

Step	Timing (US Eastern)	Action
1	15:30 ET (pre-close scanner window)	Signal adjudication: 3-Gate + Composite RS on as-of prior close / same-day bars available at T-30m (research backtests use prior close only — no look-ahead)
2	Next regular-session Open	Mechanical fills with modeled +10 bps buy / −10 bps sell slippage
3	Same bar / EOD	Proxy rebalance toward leverage target; deposit days force proxy refresh
4	Continuous	HTML-spec exits: hard stop intraday wick; trailing on close

DCA Protocol: On each **first trading day of the calendar month**, USD contribution credits cash *before* exits/entries/proxy rebalance, so new capital immediately funds open slots or the QQQ core.

3.3 Universe Construction (Anti-Survivorship)

- **PIT Dynamic-60:** Eleven sector ETFs ranked by Composite RS; quota curve fills ~60 names using only tickers already listed as of the prior close.
- IPOs enter only when listed; delisting incompleteness vs CRSP is disclosed as a residual bias.

IV. Risk Management & Exit Discipline

4.1 Hard Stop-Loss (M2 Specification)

[
P_{\text{stop}} = P_{\text{entry}} \times (1 - 0.05)
]

- Trigger: session **low** ≤ stop price → exit at `min(open, stop)` path (modeled).
- **No discretion.** Earnings gaps may fill **through** the stop (see §VI Gap Risk).

4.2 Trailing Take-Profit (Uncapped)

Activation when peak unrealized gain ≥ +15%:

```
[
\text{TrailFloor} = \max!\big(Kijun_{\{26\}},\,; PeakHigh - 2.5 \times ATR_{\{14\}}\big)
]
```

Exit on **close** below `TrailFloor` . No artificial profit cap (*Let Winners Run*).

4.3 Operational Guardrails (Live Stack)

Control	Implementation
Order idempotency	SQLite WAL mode; persistent order keys
Concurrency	ORDER_MUTEX / single-writer execution path
Slot ceiling	Hard max open names = regime slot count
Macro overlay (optional live)	MSI 2.0 risk index available in platform; C1-M2 backtest core does not cash to 50% on MSI (prior MSI-gate ablation underperformed)

V. Historical Track Record & Audit

Audit stack (all tables below): slippage 10 bps · fee 8 bps · next-open · PIT Dynamic-60 · QQQ/QLD proxy as specified.

5.1 Lump-Sum Performance — C1 Baseline vs M2 Engine

Initial simulated capital: USD 100,000 (research NAV; scale-invariant to CAGR/MDD/Sharpe).

Window A — Full Cycle (2018-08-31 → 2026-08-28)

Book	CAGR	MDD	Sharpe	Calmar	Trades	Win %	Profit Factor	vs QQQ excess CAGR
QQQ B&H	18.39%	−35.62%	0.819	0.516	—	—	—	—
QLD B&H	27.90%	−63.79%	0.752	0.437	—	—	—	—
C1 v1 (Baseline)	18.47%	−41.71%	0.725	0.443	325	24.0%	1.489	+0.08%p
M2 Engine	22.00%	−51.13%	0.761	0.430	286	28.3%	1.825	+3.61%p

Window B — AI Mega-Cycle (2023-08-31 → 2026-08-28)

Book	CAGR	MDD	Sharpe	Calmar	Trades	Win %	Profit Factor	vs QQQ excess CAGR
QQQ B&H	23.93%	−22.88%	1.152	1.046	—	—	—	—
QLD B&H	39.92%	−42.34%	1.028	0.943	—	—	—	—
C1 v1 (Baseline)	31.81%	−26.96%	1.043	1.180	129	25.6%	1.760	+7.88%p
M2 Engine	30.23%	−39.50%	0.875	0.765	128	30.5%	1.931	+6.30%p

Interpretation: M2 raises full-cycle CAGR and Profit Factor at the cost of deeper MDD and a modest AI-window CAGR trade-off versus C1. Selection between C1 and M2 is a **drawdown-budget** decision, not a capital-structure decision (§5.3).

5.2 Dollar-Cost Averaging Track Record

Cash-flow schedule: Seed **\$8,000** at (T_0) + **\$400** on each month’s first U.S. session.

Total invested (8y): \$46,400 (97 cash-flow events including seed).

Performance metric: Money-Weighted Rate of Return (**MWRR / XIRR**).

M2 configuration in this table (tuning / DCA audit): bear sleeve **2 × 40%** + hard stop **−5.0%** (identical to lump-sum M2 that printed 8y CAGR 22.00%).

Book	Total Invested	Final Equity	Net Profit	Net ROI	XIRR (MWRR)	DCA MDD	vs QQQ Profit
SPY DCA	\$46,400	\$93,009	\$46,609	100.5%	14.47%	-32.42%	-\$25,696
QQQ DCA	\$46,400	\$118,705	\$72,305	155.8%	19.49%	-28.90%	—
QLD DCA	\$46,400	\$199,134	\$152,734	329.2%	30.22%	-59.37%	+\$80,430
C1 v1 DCA	\$46,400	\$147,265	\$100,865	217.4%	23.94%	-39.99%	+\$28,561
M2 DCA	\$46,400	\$174,683	+\$128,283	+276.5%	27.48%	-48.15%	+\$55,979

M2 DCA vs QQQ DCA: approximately **+\$56,000** excess terminal wealth on identical contributions.

5.3 Capital-Structure Sensitivity (Stress)

Question tested: Does changing seed vs contribution mix flip C1 vs M2 terminal ranking?

Answer: **No.** Across all three stressed schedules, **M2 leads final equity; C1 leads DCA MDD.**

Definition note: Sensitivity runs isolate the stop change — **M2 = C1 slot/leverage identical + hard stop -5% only** (bear weights remain Baseline 2 × 25%). Terminal figures below are therefore *not* mechanically identical to §5.2's bear-40/40 M2; both audits still rank M2 first on final equity.

Scenario	Seed / Contribution	Total Invested	M2 Final Equity	C1 Final Equity	Gap (M2 – C1)
Heavy DCA	\$3,000 + \$600/mo	\$60,600	\$246,778	\$186,047	+\$60,732
Heavy Lump-sum	\$30,000 + \$300/mo	\$58,800	\$282,133	\$202,594	+\$79,539
Step-up DCA	\$8,000; \$300/mo → \$800/mo after 2022-09-01	\$60,800	\$219,158	\$173,492	+\$45,666

Investor implication: Choose **C1** if the mandate's *hard* drawdown tolerance is $\approx -40\%$; choose **M2** if the mandate prioritizes terminal wealth and accepts $\approx -45\%$ to -51% path MDD.

5.4 Rejected / Ablated Designs (Transparency)

The following were tested under identical friction and **permanently discarded** for the production C1–M2 engine:

Ablation	Result	Decision
SMA200 50% cash sleeve	Underperformed QQQ B&H	Rejected
High-hurdle solo satellite (+10pt RS)	8y CAGR ~7%	Rejected
Weekly 3–5d mean-reversion	Negative CAGR	Rejected
PEAD (+15% surprise, pure 60d hold)	Lost to C1 both windows	PEAD research closed
MSI climate gate (stocks → QQQ/cash)	Cut V-recoveries; 8y ~2%	Rejected as C1 replacement
Entry squeeze ($k_{ijun} \leq 3.5\%$ + ATR contraction)	Over-filtered leaders; CAGR collapsed	Rejected
Early trail (+12%) / Wide trail (+18%, ATR 3)	Destroyed or diluted right tail	Rejected
Pure S&P timing (SSO/UPRO/SH switches)	All lost to C1 on 8y CAGR	Index timing not a substitute

VI. Risk Factors & Disclosures

6.1 Market Risk

Nasdaq and S&P drawdowns can and will produce **losses of invested capital**. Simulated peak-to-trough declines of approximately **-40% (C1)** to **-51% (M2 lump-sum)** have already been observed in-sample.

6.2 Leverage & Volatility Decay

The 1.5x module (QQQ+QLD) introduces **path-dependent decay** in choppy bull regimes even when the spot index is flat-to-up. Gross exposure is cash-constrained but still amplifies left-tail events while QLD is held.

6.3 Execution, Gap & Model Risk

- Pre-market earnings/news gaps may fill **beyond** the -5% hard stop.
- Slippage may exceed the audited 10 bps in stress liquidity.
- PIT Dynamic-60 is **not** CRSP-complete; missing bankrupt/acquired names bias results.
- Parameter choices (stop -5%, trail +15%, RS weights) were researched in-sample; **overfitting risk remains**.

6.4 Liquidity & Concentration

Three-name conviction sizing implies **issuer concentration**. A single leadership name at 50% NAV can dominate P&L.

6.5 Operational & Technology Risk

Live deployment depends on Korea Investment & Securities (KIS) API availability, process uptime of PortfolioGuardian, and correct clock synchronization to US cash equity sessions.

6.6 Past Performance Disclaimer

PAST SIMULATED PERFORMANCE IS NOT INDICATIVE OF FUTURE RESULTS.
No representation is made that any account will or is likely to achieve profits or losses similar to those shown. Hypothetical results have inherent limitations, including the benefit of hindsight in rule selection. Live trading involves risk of loss **greater than or equal to** the entire amount invested.

6.7 No Offer; Eligibility

This PPM/KIID-style memorandum does **not** constitute an offer to sell or a solicitation to buy any security or fund interest in any jurisdiction where such offer would be unlawful. Interests, if any, would be offered only via definitive subscription documents to eligible investors.

VII. Technology & Infrastructure Specification

Layer	Specification
Language / Runtime	Python 3.11+
API Surface	FastAPI service plane (server.py)
State / Orders	SQLite 3 with WAL mode; idempotent order keys
Broker Connectivity	Korea Investment & Securities (KIS) OpenAPI + WebSocket market data
Risk Daemon	24/7 background monitor — PortfolioGuardian (stop/trail/slot enforcement)
Research SSOT	research_and_backtests/ engines: qqq_beat_alpha_backtester.py (C1), c1_v1_tuning.py (M2 controls), c1_v1_dca_simulation.py , dca_capital_sensitivity.py
Artifact Audit Trail	JSON caches under data/backtest_cache/ (c1_v1_tuning_results.json , c1_v1_dca_results.json , dca_capital_sensitivity_results.json)

7.1 High-Water Mark (Conceptual Reporting)

For investor reporting, strategy NAV high-water mark (HWM) is defined as:

```
[
HWM_t = \max_{0 \le s \le t} NAV_s
]
```

Drawdown at (t):

[
DD_t = \frac{NAV_t}{HWM_t} - 1
]

DCA mandates additionally report **invested-capital ratio** ($NAV_t / \sum \text{Contributions}_{0..t}$) and its drawdown (see research artifacts: `invested_ratio_mdd_pct`).

VIII. Appendix — Glossary

Term	Meaning
Alpha	Excess return vs a stated benchmark after beta adjustment (here also shown as excess CAGR and Jensen α in research JSON)
Asymmetric Payoff	Limited loss per trade (−5% stop) vs uncapped gain via trailing
Calmar Ratio	CAGR / MDD
Composite RS	Multi-horizon relative strength blend (21/63/126d)
DCA	Dollar-cost averaging — fixed contributions on a schedule
HWM	High-Water Mark
KIID	Key Investor Information Document (UCITS-style factsheet format)
MDD	Maximum peak-to-trough drawdown
MWRR / XIRR	Money-weighted rate of return for irregular cash flows
PIT	Point-in-Time universe — no future listings
PPM	Private Placement Memorandum
Sharpe Ratio	Annualized mean excess return / volatility (rf ≈ 0 in research tables)

IX. Document Control

Version	Date	Authoring Context	Change
1.0	2026-09-08	AI-Sangmoo Quant Research / Cloud Agent audit chain	Initial C1-M2 PPM incorporating lump-sum, DCA, and capital-sensitivity audits

End of Memorandum